Spatial–Spectral–Temporal Correlation Filter for Hyperspectral Object Tracking

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***Abstract*— Object tracking with hyperspectral videos (HSVs) offers significant advantages due to the captured spectral fingerprint information, which provides detailed physical material characteristics. While correlation filter (CF)-based tracking methods align well with the high-dimensional nature of HSVs, they often fall short of fully utilizing the spatial–spectral–temporal structure inherent in these data. In this article, we introduce a spatial–spectral–temporal CF (SSTCF) framework to address these limitations. SSTCF employs the spatial–spectral histogram of gradients and fractional abundances as features to characterize the spatial–spectral structure of the object. A low-rank constraint is integrated into the CF framework to enhance the global spectral semantic dependencies among learned filters. In addition, a temporal constraint is incorporated to ensure filter consistency across consecutive frames, further improving tracking continuity between nearby frames. Extensive experiments demonstrate that our SSTCF tracker achieves more accurate and stable performance. The source code will be publicly available at https://github.com/bearshng/SSTCF**

***Index Terms*— Alternating direction method of multiplier (ADMM), correlation filter (CF), hyperspectral object tracking, spatial–spectral–temporal modeling.**

I. Introduction

OBJECT tracking is a fundamental task in both computer vision and remote sensing, with wide-ranging applications in video surveillance, military security, and Earth observation [1], [2], [3], [4], [5]. Traditional color video tracking primarily captures visible spatial structural information but often neglects underlying physical properties, such as material composition. Consequently, tracking with color videos can be challenging in scenarios involving background clutter (BC) or dim, small objects [6], where spatial structural information alone is insufficient. Hyperspectral videos (HSVs), on the other hand, provide rich spectral fingerprint information that reflects the physical material properties of objects, offering significant advantages for object tracking. This has led to increasing interest in hyperspectral object tracking in recent years [7], [8], [9].

Building an effective tracking framework to locate the object is a crucial step in hyperspectral object tracking. Traditional methods such as the Kalman filter and the mean-shift algorithm have been employed for tracking in HSVs, as seen in [10] and [11], respectively. Correlation filter (CF)-based trackers have demonstrated greater robustness in challenging tracking scenarios [12], [13], [14] and can be efficiently computed in the Fourier domain by enforcing the data to follow a circular structure [15]. This efficiency, combined with the ability to handle variations in object appearance, makes CF-based methods particularly well-suited for the high-dimensional nature of hyperspectral data [16], [17]. Kernelized CFs were adopted in [18], [19], and [20] to quickly distinguish the hyperspectral targets from the neighboring environment. In order to suppress the boundary effect of CF and unexpected peaks in the response map, sparse constraint and context-aware constraint terms are introduced in [21]. MHT [8] leverages the background-aware CF (BACF) [22], which introduces a predefined mask to distinguish the target from the background and employs the spatial–spectral histogram of multidimensional gradients (SSHMGs) and fractional abundances as features for tracking.

However, MHT suffers from two significant drawbacks. First, MHT uses SSHMG as the feature descriptor to represent the object. Due to its local nature, SSHMG fails to capture the global relationships among spectral bands. This lack of global correlation limits the effectiveness of the representation and decreases the discriminative ability of the learned filters. Second, MHT lacks temporal consistency between nearby frames, reducing its robustness, particularly in scenarios involving occlusions, BC, and sudden scene changes.

To address the limitations of MHT, this article introduces a novel spatial–spectral–temporal CF (SSTCF) framework for hyperspectral object tracking. The key innovation of SSTCF lies in its ability to integrate global spectral correlations within a frame and temporal consistency between frames. Given the low-dimensional nature of hyperspectral images, the extracted SSHMG features are highly correlated and tend to reside in a low-dimensional subspace. As presented in Fig. 1, the SSTCF framework incorporates a low-rank constraint to the learned filters to exploit the low-dimensional structure of hyperspectral data, which compensates for the locality of the SSHMG

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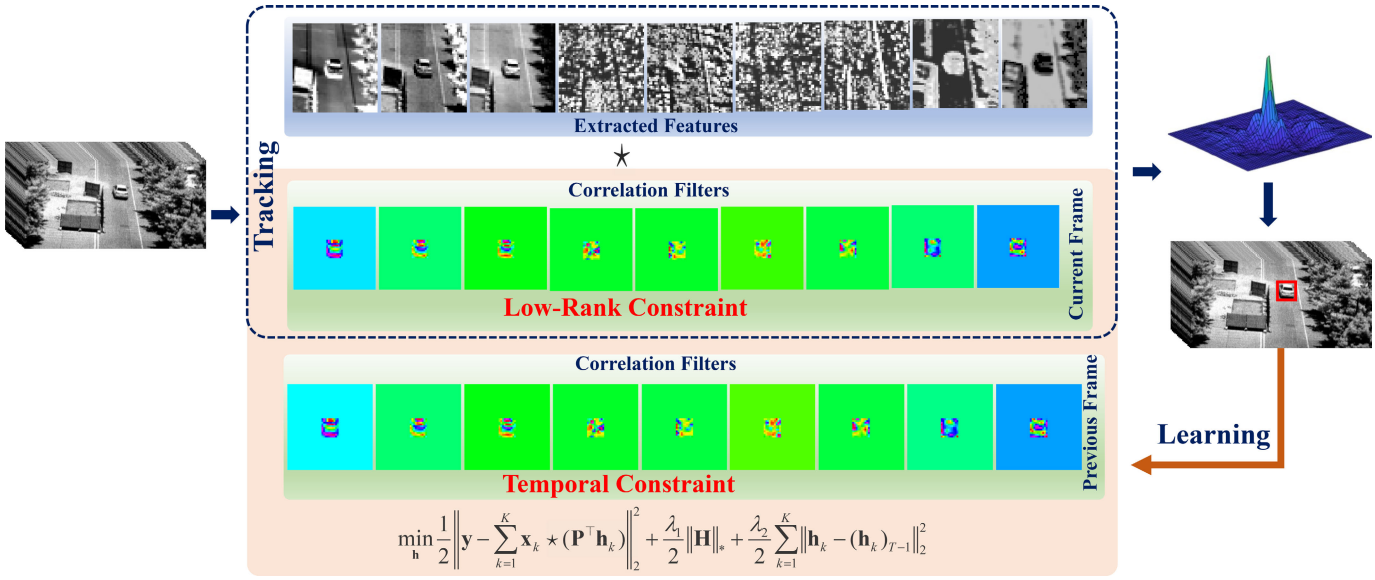


Fig. 1. Framework of our SSTCF tracker incorporates a low-rank constraint on the filters to enforce global correlation among them, along with a temporal constraint to ensure continuity in tracking between adjacent frames.

descriptor. This low-rank representation also ensures that the extracted SSHMG and abundance features interact effectively so that they can be jointly utilized to learn more discriminative filters, improving the overall robustness of the tracking system. In addition, SSTCF imposes a temporal constraint on the learned filters to ensure tracking consistency across consecutive frames, making the tracker more resilient to sudden changes in the target or scene, such as occlusions or BC. The corresponding optimization problem is efficiently solved using the alternating direction method of multipliers (ADMMs).

The contribution of this article can be summarized as follows.

1) A novel spatial–spectral–temporal CF framework is introduced for robust hyperspectral object tracking. To the best of our knowledge, this is the first time that global spectral correlation has been considered within the CF framework.
2) An effective optimization method-based ADMM algorithm is developed to learn the filters, preserving the computational efficiency of the original CF.

The remainder of this article is organized as follows. Section II reviews the recent works in hyperspectral tracking and low-dimensional modeling of HSIs. Section III describes the background of introducing SSTCF. Section IV introduces the details of the proposed SSTCF. Section V presents quantitative and visual experiments. Finally, Section VI concludes this article.

II. Related Work

This section describes the recent advances in hyperspectral object tracking and low-dimensional representation of HSIs.

A. Hyperspectral Object Tracking

Feature extraction is a crucial first step in hyperspectral tracking. HSVs encompass spatial, spectral, and temporal information, enabling feature extraction along individual dimensions or through their combination. Banerjee et al. [23] used captured spectra as features and estimated the real reflectance of objects to achieve illumination-invariant tracking. Instead of relying on raw spectral data, Uzkent et al. [10] enhanced tracking robustness by computing local hyperspectral histograms, making the tracker more resilient to varying illumination conditions. Henrot et al. [24] performed unmixing on HSVs to obtain finer spectral descriptions of scenes, particularly effective for gas plume tracking. For spatial feature extraction, Hou et al. [21] employed the histogram of oriented gradients (HOGs) feature extractor in a band-by-band manner and concatenated it with intensity HOG for representation. Moving beyond separate spatial and spectral feature extraction, the spectral–spatial histogram of multidimensional gradients (SSHMG) [25] calculated joint spatial–spectral gradient orientations to encode the 3-D local spectral–spatial structure in HSVs. By combining these features with fractional abundances, this method significantly enhanced tracking robustness. Qian et al. [18] extracted 3-D patches as fixed feature descriptors to model the visual appearance of targets. Instead of using handcrafted features, deep learning-based approaches further improve tracking performance due to their powerful representational abilities [26], [27], [28], [29]. For example, in [26], [30], [31], and [32], hyperspectral frames were divided into multiple groups of false-color images, enabling the use of pretrained backbones designed for color images to achieve more robust representations. Consequently, tracking performance and robustness were significantly improved. Gao et al. [7] and [33] leveraged CNNs and Transformers to depict global spectral correlations, enhancing discriminative ability. Wang et al. [34] introduced a Transformer-based context fusion module and hyperspectral cross-attention to capture self-context and cross-context feature interactions, leading to more robust interactions between the template and the search region.

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B. Low-Dimensional Representation of HSIs

Effectively capturing the low-dimensional structure embedded in original data is highly beneficial for pattern recognition, as demonstrated in face recognition [35], hyperspectral classification [36], [37], [38], and hyperspectral unmixing [39], [40]. The low-dimensional representation of HSIs can be described in terms of sparsity or low rankness. The sparse representation refers to encoding the original signals using a linear combination of fewer basic signals. For instance, Liu et al. [37] developed a joint spectral–spatial sparse coding model that enforces consistent coding coefficients at the same spatial location across different bands, effectively preserving global spatial structures and significantly enhancing discriminative power. Similarly, Lu et al. [39] incorporated manifold regularization into abundance estimation to maintain a strong connection between the original image and the material abundance maps, resulting in improved unmixing performance. The low-rank representation captures the low-dimensional structure within data by modeling the entire dataset as lying in a low-dimensional subspace. This property makes it highly effective in noise reduction [41] and anomaly detection [42]. In this article, we employ low-rank representation to capture the correlation between extracted SSHMG features and abundances, leveraging its ability to ensure global consistency and robustness to noise.

III. Background

In this section, we introduce the preliminaries of CFs and the MHT tracker.

A. Correlation Filters

CFs aim to produce higher responses when two signals are similar while yielding low responses for dissimilar signals. Let $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_K] \in \mathbb{R}^{N\times K}$ represent the extracted multichannel feature and $\mathbf{H} = [\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_K] \in \mathbb{R}^{D\times K}$ represent the CF, where N, K, and D denote the number of pixels, the dimensionality of extracted features, and the size of the CF, respectively. The score can be calculated in the spatial domain by performing convolution between $\mathbf{x}$ and $\mathbf{h}$. Instead of computing the correlation in the spatial domain, this process can be efficiently handled in the frequency domain using elementwise multiplications. Denoting the Fourier transform of a vector with the symbol $\hat{}$, the correlation filtering process can be mathematically expressed as

$$\sum_{k=1}^{K} \mathbf{x}_k \star \mathbf{h}_k = \sum_{k=1}^{K} \mathcal{F}^{-1}\left(\hat{\mathbf{x}}_k \odot \hat{\mathbf{h}}_k^*\right) \tag{1}$$

where $\mathcal{F}^{-1}$ denotes the inverse fast Fourier transform (FFT), $\odot$ is the elementwise multiplication, $*$ is the complex conjugate, and $\star$ is the spatial correlation operator.

Learning the CF amounts to minimizing the following objective function:

$$\min_{\mathbf{h}} \frac{1}{2}\|\mathbf{y} - \sum_{k=1}^{K} \mathbf{x}_k \star \mathbf{h}_k\|_2^2 + \frac{\lambda}{2}\sum_{k=1}^{K}\|\mathbf{h}_k\|_2^2 \tag{2}$$

where $\mathbf{y} \in \mathbb{R}^{N\times 1}$ is the desired response with Gaussian shape and λ is a regularization parameter to avoid dividing by zero.

Hyperspectral images are high-dimensional in both spectral and spatial domains, leading to high computational demands. However, CFs use a cyclic shift structure in the training samples, allowing spatial-domain convolutions to be transformed into elementwise multiplications in the frequency domain. Given an image of size $n \times n$, the complexity of circular convolution in the spatial domain is $O(n^4)$. By contrast, employing FFT for elementwise multiplications reduces this to $O(n^2 \log n)$, significantly decreasing computational complexity. For this reason, we develop our hyperspectral tracker in the framework of CF.

B. Revisiting MHT

In the original CF-based tracking, the model relies on a single positive example, with negative examples generated by applying a circular shift to the positive example. As a result, the CF-based tracking is vulnerable to overfilling. The MHT employs a background-aware CF framework [22], which introduces a spatial constraint on the CF to mitigate unwanted boundary effects. Specifically, a binary mask $\mathbf{P} \in \mathbb{R}^{D\times N}$ is applied to $\mathbf{h}$ to generate the true positive and negative samples, producing the following objective function:

$$\min_{\mathbf{h}} \frac{1}{2}\|\mathbf{y} - \sum_{k=1}^{K} \mathbf{x}_k \star (\mathbf{P}^\top \mathbf{h}_k)\|_2^2 + \frac{\lambda}{2}\sum_{k=1}^{K}\|\mathbf{h}_k\|_2^2. \tag{3}$$

To more effectively model the visual appearance of the hyperspectral object, MHT employs the SSHMG and fractional abundances as features. SSHMG captures the local spatial–spectral texture by computing the histogram of spatial and spectral gradient orientations, while fractional abundances encode the global distribution of underlying materials.

However, the MHT suffers from two drawbacks. First, SSHMG only describes the local spatial–spectral texture of the object, overlooking the global spectral correlation among bands. Given the proven advantages of global features in pattern recognition [43], [44], [45], neglecting this global spectral correlation leads to suboptimal tracking performance. Second, MHT fails to account for the temporal continuity between nearby frames, increasing the risk of losing the object when unexpected distractors appear. In this article, we address these limitations by developing a spatial–spectral–temporal CF that effectively captures both the global spectral correlation and maintains the temporal consistency.

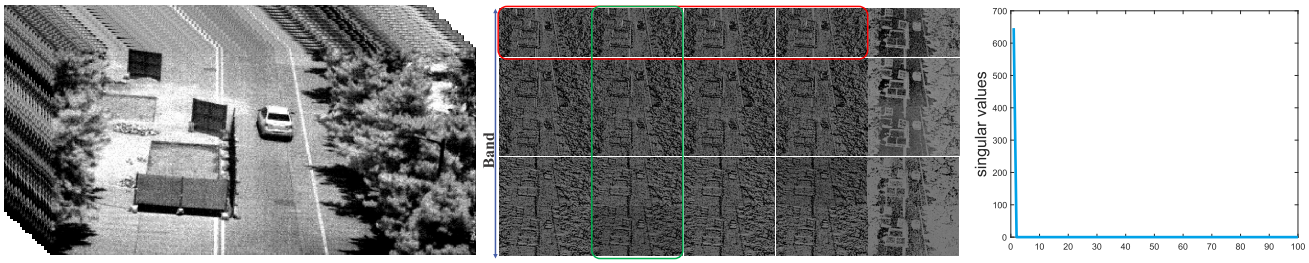
IV. Spatial–Spectral–Temporal CF

In this section, we introduce the details of our spatial–spectral–temporal CF and also its optimization.

A. Spatial–Spectral–Temporal CF

In order to address the aforementioned challenges, we impose a low-rank constraint and temporal constraint on the filters $\mathbf{h}$. Given the global spectral correlation inherent in hyperspectral images, the extracted SSHMG features are highly coupled, meaning that there is significant similarity

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(a) (b) (c)

Fig. 2. Demonastration of the low-rank structure of SSHMG. (a) Typical HSI. (b) Extracted SSHMG features where the red part denotes the typical features extracted within individual bands, while the green part represents the corresponding features extracted across different bands. (c) Curve of singular values (only the top 100 values are plotted).

among them. To illustrate this, we present typical features in Fig. 2(b). It can be observed that there are very obvious similarities within bands and between bands. We further plot the corresponding singular value curves in Fig. 2(c). For better illustration, we display only the top 100 singular values. As observed, there is a clear decaying trend in the singular values. Such a phenomenon also evidently shows that there exists a low-dimensional structure among the extracted SSHMG feature. The low-dimensional structure can be mathematically depicted by sparse or low-rank representation. In this article, we resort to the use of the low-rank representation.

To ensure temporal consistency, we also enforce the learned CF to be similar across nearby frames. We fulfill this goal by minimizing the distance between the learned filters in adjacent frames. Incorporating both the low-rank and temporal constraints into (3), our SSTCF model at the Tth frame is learned by minimizing the following objective function:

$$\min_{\mathbf{h}} \frac{1}{2}\|\mathbf{y} - \sum_{k=1}^{K} \mathbf{x}_k \star (\mathbf{P}^\top \mathbf{h}_k)\|_2^2 + \frac{\lambda_1}{2}\|\mathbf{H}\|_* + \frac{\lambda_2}{2}\sum_{k=1}^{K}\|\mathbf{h}_k - (\mathbf{h}_k)_{T-1}\|_2^2 \tag{4}$$

where the first term ensures the data fidelity, the second one enables the low rankness of learned filters, and the third cone ensures the temporal consistency. λ_1 and λ_2 are two hyperparameters to balance three terms. The nuclear norm $\|\cdot\|_*$ is defined as

$$\|\mathbf{H}\|_* = \sum_i \sigma_i(\mathbf{H}) \tag{5}$$

where $\sigma_i(\mathbf{H})$ represents the ith singular value of $\mathbf{H}$. We apply the nuclear norm to all extracted features to not only capture the global spectral correlation but also enable the joint learning of the filters of SSHMG and fractional abundances, effectively exploiting the relationships among different features.

B. Optimization of Our SSTCF Model

Inspired by the previous works, we learn the CFs in the frequency domain for efficient computation. In the frequency domain, (4) can equivalently be formulated as

$$\min_{\{\mathbf{h},\hat{\mathbf{g}}_k\}} \frac{1}{2}\|\hat{\mathbf{y}} - \hat{\mathbf{X}}\hat{\mathbf{g}}\|_2^2 + \frac{\lambda_1}{2}\|\mathbf{H}\|_* + \frac{\lambda_2}{2}\sum_{k=1}^{K}\|\hat{\mathbf{g}}_k - (\hat{\mathbf{g}}_k)^{T-1}\|_2^2 \quad \text{s.t. } \hat{\mathbf{g}}_k = \sqrt{N}\mathbf{F}\mathbf{P}^\top\mathbf{h}_k \tag{6}$$

where the matrix $\hat{\mathbf{X}} = [\text{diag}(\hat{\mathbf{x}}_1)^\top, \ldots, \text{diag}(\hat{\mathbf{x}}_K)^\top]$ of size $N \times KN$ and $\mathbf{g}$ is an auxiliary variable. Here, $\mathbf{F}$ is an $N \times N$ orthonormal matrix of complex basis vectors to map any N dimensional vectorized signal to the Fourier domain, mathematically formulated as $\hat{\mathbf{a}} = \sqrt{N}\mathbf{F}\mathbf{a}$, where $\mathbf{a} \in \mathbb{R}^{N\times 1}$. The transpose operator $\top$ on a complex vector or matrix computes the conjugate transpose.

Under the framework of ADMM, the augmented Lagrangian form of (6) can be expressed as

$$\mathcal{L}(\mathbf{H}, \hat{\mathbf{g}}, \Gamma) = \frac{1}{2}\|\hat{\mathbf{y}} - \hat{\mathbf{X}}\hat{\mathbf{g}}\|_2^2 + \frac{\lambda_1}{2}\|\mathbf{H}\|_* + \frac{\lambda_2}{2}\sum_{k=1}^{K}\|\hat{\mathbf{g}}_k - (\hat{\mathbf{g}}_k)_{T-1}\|_2^2 + \sum_{k=1}^{K}\left\langle \hat{\gamma}_k^\top, \hat{\mathbf{g}}_k - \sqrt{N}\mathbf{F}\mathbf{P}^\top\mathbf{h}_k\right\rangle + \frac{\mu}{2}\sum_{k=1}^{K}\|\hat{\mathbf{g}}_k - \sqrt{N}\mathbf{F}\mathbf{P}^\top\mathbf{h}_k\|_2^2 \tag{7}$$

where $\Gamma = [\gamma_1, \gamma_2, \ldots, \gamma_K] \in \mathbb{R}^{N\times K}$ is the Lagrange multiplier, μ is the stepsize, and $\hat{\Gamma} = [\hat{\gamma}_1, \hat{\gamma}_2, \ldots, \hat{\gamma}_K]$ is the counterpart in the Frequency domain. We now solve each variable in an alternating manner.

1) Subproblem $\mathbf{H}^*$: Extracting all the items related to $\mathbf{H}^*$, we can obtain the following subproblem:

$$\begin{aligned}\mathbf{H}^* &= \arg\min_{\mathbf{H}} = \frac{\lambda_1}{2}\|\mathbf{H}\|_* + \sum_{k=1}^{K}\left\langle \hat{\gamma}_k^\top, \hat{\mathbf{g}}_k - \sqrt{N}\mathbf{F}\mathbf{P}^\top\mathbf{h}_k\right\rangle + \frac{\mu}{2}\sum_{k=1}^{K}\|\hat{\mathbf{g}}_k - \sqrt{N}\mathbf{F}\mathbf{P}^\top\mathbf{h}_k\|_2^2 \\ &= \frac{\lambda_1}{2}\|\mathbf{H}\|_* + \frac{\mu}{2}\sum_{k=1}^{K}\|\hat{\mathbf{g}}_k - \sqrt{N}\mathbf{F}\mathbf{P}^\top\mathbf{h}_k + \hat{\gamma}_k^\top/\mu\|_2^2 \\ &= \frac{\lambda_1}{2}\|\mathbf{H}\|_* + \frac{\mu}{2}\|\widehat{\mathbf{G}} - \sqrt{N}\mathbf{F}\mathbf{P}^\top\mathbf{H} + \widehat{\Gamma}^\top/\mu\|_2^2. \end{aligned} \tag{8}$$

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Here, $\mathbf{G}$ and Γ are the inverse FFT (IFFT) of $\widehat{\mathbf{G}}$ and $\widehat{\Gamma}$, and their computation can be divided into K independent IFFT of $\mathbf{g}_k = (1/\sqrt{N})\mathbf{P}\mathbf{F}^\top \hat{\mathbf{g}}_k$ and $\gamma_k = (1/\sqrt{N})\mathbf{P}\mathbf{F}^\top \hat{\gamma}_k$.

Given a matrix $\mathbf{X} \in \mathbb{R}^{n_1 \times n_2}$, we define the soft-thresholding operator $\mathcal{D}_\tau$ as follows:

$$\mathcal{D}_\tau(\mathbf{X}) = \mathbf{U}\mathcal{D}_\tau(\Sigma)\mathbf{V}^\top, \quad \mathcal{D}_\tau(\Sigma) = \operatorname{diag}\big(\{\sigma_i - \tau\}_+\big) \tag{9}$$

where $\mathbf{U}$, $\mathbf{V}$, and Σ are the left singular matrix, the right singular matrix, and the singular value matrix of $\mathbf{X}$, respectively. Using $\mathcal{D}_\tau$, the solution of $\mathbf{H}$ can be formulated as

$$\mathbf{H}^* = \mathcal{D}_{\lambda_1/\mu}(\mathbf{P}(\mathbf{G} + \Gamma/\mu)). \tag{10}$$

In our implementation, we calculate $\mathbf{g}_k$ and γ_k by applying IFFT on each $\hat{\mathbf{g}}_k$ and $\hat{\gamma}_k$ and then applying the lookup table formed from the masking matrix $\mathbf{P}$.

2) Subproblem $\mathbf{g}^*$*:* Fixing $\mathbf{H}$ and Γ, solving $\mathbf{g}^*$ reduces to the following problem:

$$\begin{aligned}\hat{\mathbf{g}}^* = \arg\min_{\hat{\mathbf{g}}} &\frac{1}{2}\|\hat{\mathbf{y}} - \hat{\mathbf{X}}\hat{\mathbf{g}}\|_2^2 + \frac{\lambda_2}{2}\sum_{k=1}^{K}\|\hat{\mathbf{g}}_k - (\hat{\mathbf{g}}_k)_{T-1}\|_2^2 \\ &+ \frac{\mu}{2}\sum_{k=1}^{K}\|\hat{\mathbf{g}}_k - \sqrt{N}\mathbf{F}\mathbf{P}^\top\mathbf{h}_k + \hat{\gamma}_k^\top/\mu\|_2^2.\end{aligned} \tag{11}$$

Equation (11) can be alternatively written as

$$\begin{aligned}\hat{\mathbf{g}}^* = \arg\min_{\hat{\mathbf{g}}} &\frac{1}{2}\sum_{n=1}^{N}\|\hat{\mathbf{y}}(n) - \hat{\mathbf{x}}(n)\hat{\mathbf{g}}(n)\|_2^2 \\ &+ \frac{\lambda_2}{2}\sum_{n=1}^{N}\|\hat{\mathbf{g}}(n) - (\hat{\mathbf{g}}(n))_{T-1}\|_2^2 \\ &+ \frac{\mu}{2}\sum_{t=1}^{N}\|\hat{\mathbf{g}}(n) - \sqrt{N}\mathbf{F}\mathbf{P}^\top\mathbf{h}(n) + \hat{\gamma}(n)^\top/\mu\|_2^2.\end{aligned} \tag{12}$$

As can observed, each element of $\hat{\mathbf{y}}$ only depends on K values of $\hat{\mathbf{x}}_n = [\hat{\mathbf{x}}_1(n), \ldots, \hat{\mathbf{x}}_K(n)]^\top$ and $\hat{\mathbf{g}}^n = [\operatorname{conj}(\hat{\mathbf{g}}_1(n)), \ldots, \hat{\mathbf{g}}_K(n)]^\top$, where conj computes the complex conjugate. Therefore, this problem can be decomposed into T subproblem with each being

$$\begin{aligned}\arg\min_{\hat{\mathbf{g}}(n)} = &\frac{1}{2}\|\hat{\mathbf{y}}(n) - \hat{\mathbf{x}}(n)^\top\hat{\mathbf{g}}(n)\|_2^2 + \frac{\lambda_2}{2}\|\hat{\mathbf{g}}(n) - \hat{\mathbf{g}}(n)_{T-1}\|_2^2 \\ &+ \frac{\mu}{2}\|\hat{\mathbf{g}}(n) - \sqrt{N}\mathbf{F}\mathbf{P}^\top\mathbf{h}(n) + \hat{\gamma}(n)^\top/\mu\|_2^2.\end{aligned} \tag{13}$$

Setting the derivative with respect to $\hat{\mathbf{g}}(n)$ to zero, the closed-form solution can be obtained by

$$\begin{aligned}\hat{\mathbf{g}}(n)^* = &\big(\hat{\mathbf{x}}(n)\hat{\mathbf{x}}(n)^\top + (\lambda_2 + \mu)N\mathbf{I}\big)^{-1} \\ &\times \big(\hat{\mathbf{x}}(n)\hat{\mathbf{y}}(n) + \lambda_2 N(\hat{\mathbf{g}}(n))_{T-1} + \mu N\hat{\mathbf{h}}(n) - \mu\hat{\gamma}(n)\big)\end{aligned} \tag{14}$$

where $\mathbf{I}$ is an identity matrix. In order to meet the requirement of real-time tracking, we compute the $(\hat{\mathbf{x}}(n)\hat{\mathbf{x}}(n)^\top + (\lambda_2 + \mu)N\mathbf{I})^{-1}$ term using the Sherman–Morrison formula such that $(\mathbf{A} + \mathbf{u}\mathbf{v}^\top)^{-1} = \mathbf{A}^{-1} - (\mathbf{A}^{-1}\mathbf{u}\mathbf{v}^\top\mathbf{A}^{-1})(1 + \mathbf{v}^\top\mathbf{A}^{-1}\mathbf{u})$ where $\mathbf{u}$ and $\mathbf{v}$ are two column vectors. Accordingly, (14) can be reformulated as

$$\begin{aligned}\hat{\mathbf{g}}(n)^* = &\frac{1}{\lambda_2 + \mu}\Big(\hat{\mathbf{x}}(n)\hat{\mathbf{y}}(n)/N + \lambda_2\big(\hat{\mathbf{g}}(n)\big)_{T-1} \\ &+ \mu\sqrt{N}\mathbf{F}\mathbf{P}^\top\mathbf{h}(n) - \hat{\gamma}(n)\Big) \\ &- \frac{\hat{\mathbf{x}}(n)}{\lambda_2 + \mu}\big(N\hat{\mathbf{y}}(n)\hat{\mathbf{x}}(n)^\top\hat{\mathbf{x}}(n) \\ &+ \hat{\mathbf{x}}(n)^\top\hat{\gamma}(n) + \mu\hat{\mathbf{x}}(n)^\top\hat{\mathbf{h}}(n) + \lambda_2\hat{\mathbf{x}}(n)^\top\big(\hat{\mathbf{g}}(n)\big)_{T-1}\Big).\end{aligned} \tag{15}$$

3) Lagrangian Update: We update the Lagrangians as

$$\gamma^{(i+1)} = \gamma^{(i)} + \mu * \big(\mathbf{g}^{(i+1)} - \mathbf{h}^{(i+1)}\big) \tag{16}$$

where i indexes the iteration.

4) Updating the Stepsize μ*:* We gradually increase μ by

$$\mu^{(i+1)} = \min(10^5, 10\mu) \tag{17}$$

where the initial μ is 1. The optimization of SSTCF is summarized in Algorithm 1. Each subproblem within SSTCF is convex, ensuring convergence as per the Eckstein–Bertsekas condition [46]. For computational efficiency, the number of iterations is empirically set to 5 in our experiments. In addition, to avoid unexpected results in the first frame where all elements of $\mathbf{g}$ are initially zero, we set $\gamma_2 = 0$.

Algorithm 1 Optimization of Our SSTCF Tracker

Input: Training Data: $\mathbf{X}$ and $\mathbf{y}$; The hyperparameter λ_1, λ_2 and μ.

1: Initialize $\Gamma = 0$;

Output: Correlation filters $\mathbf{H} = [\mathbf{h}_1, \ldots, \mathbf{h}_K]$.

2: **while** *not converged* **do**
3: Update $\mathbf{g}$ with Eq. (15);
4: Update $\mathbf{h}$ with Eq. (10);
5: Update γ with Eq. (16);
6: Update μ with Eq. (17);
7: **end while**

C. Online Tracking

In our approach, we adhere to the tracking-by-detection framework, where the correlation response map is computed via (1). To handle scale changes in the target, we apply the CF $\mathbf{h}$ across multiple resolutions of the search area. This process generates S correlation response maps, corresponding to the number of scales considered. The locational of the target is determined by identifying the position with the maximum response across these maps.

The computation of (10) primarily involves linear algebra operations and singular value decomposition. For the former, the computation is bounded by $\mathcal{O}(KN\log(N))$, where K is the number of channels and $N\log(N)$ represents the cost of computing the IFFT of a signal of length N. For the latter, the computation complexity is bounded by $\mathcal{O}(\min(KN^2, N^2K))$. According to [22], the complexity of (15) is $\mathcal{O}(NK^3)$. The computation of (16) requires $\mathcal{O}(KN)$. Therefore, the total computational complexity of SSTCF is $\mathcal{O}(\min(KN^2, N^2K)) + \mathcal{O}(NK^3)$.

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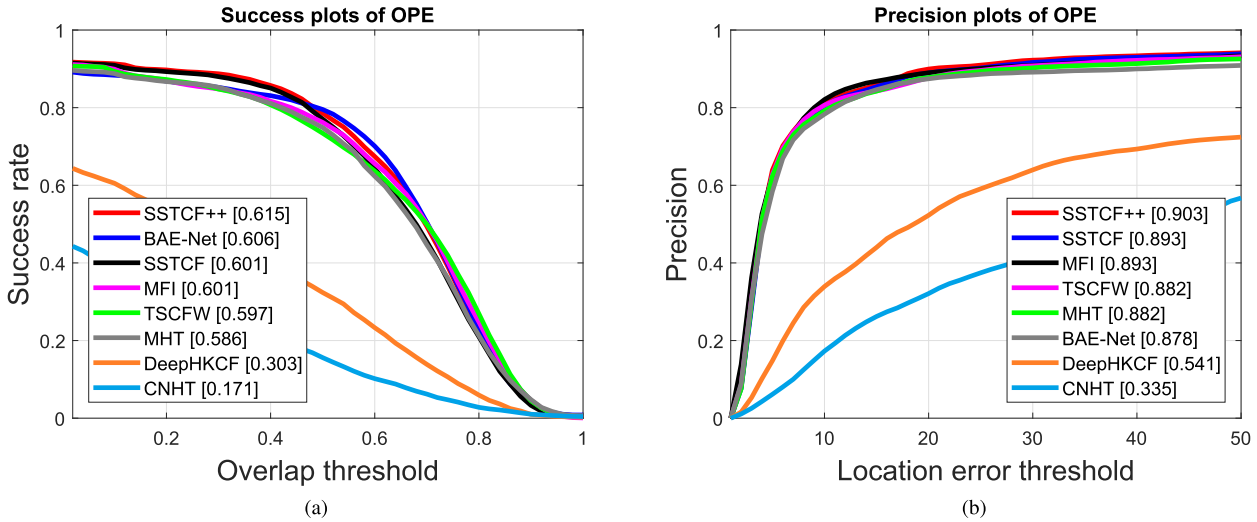


Fig. 3. Success and precision plots of all the compared hyperspectral trackers on the HOT2020 dataset. (a) Success plot. (b) Precision plot.

V. Experiments and Analysis

In this section, we conduct extensive experiments with alternative trackers to show the robustness of our SSTCF tracker.

A. Experiment Setup

1) Implementation Details: The SSTCF tracker is implemented using MATLAB 2018a. We run our method on a Linux machine equipped with an Intel[1] Xeon Silver 4214R CPU (2.40 GHz) and 126 GB of RAM. The parameters for SSHMG and abundances are consistent with those used in MHT. For scale estimation, the number of scales S is set to 5, and the scale factor is set to 1.01.

2) Experimental Data: The performance of the SSTCF tracker is evaluated using the HOT2020 and IMEC25 datasets, which are configured as follows:

1) *HOT2020 Dataset:* The HOT2020 dataset is the first large-scale dataset for hyperspectral object tracking purposes and was acquired by Xiong et al. [25] at a frame rate of 25. The dataset contains 75 videos, with 40 for training and 35 for testing. Each video contains 16 bands in the visible range from 460 to 600 nm and is labeled with 11 challenging attributes for attribute-based evaluation. These attributes include fast motion (FM), BC, illumination variation (IV), out-of-plane rotation (OPR), occlusion (OCC), motion blur (MB), low resolution (LR), scale variation (SV), in-plane rotation (IPR), out-of-view (OV), and deformation (DEF).
2) *IMEC25 Dataset:* The IMEC25 dataset was collected by Chen et al. [47] at a frame rate of 10 and includes 135 annotated sequences, with 55 used for training and 80 for testing. Each video contains 25 bands ranging from 680 to 960 nm. Similar to the HOT2020 dataset, all videos are labeled with the same 11 challenging attributes.

[1]Registered trademark.

3) Evaluation Index: Tracking performance is quantified using precision and success plots. The precision plot measures the ratio of frames where the center location error between the predicted and ground-truth bounding boxes is below a given threshold. In our experiment, we set the threshold from 1 to 50 pixels. The success plot calculates the percentage of frames where the overlap ratio between the predicted and ground-truth bounding boxes exceeds a given threshold. In our experiment, the thresholds range from 0 to 1. The trackers are ranked based on the distance precision at 20 pixels and the area under the curve (AUC) of the success plot.

4) Compared Methods: Seven methods are compared, including the baseline MHT [25], CNHT [18], TSCFW [21], MFI [20], DeepHKCF [48], SiamBAG [31], and BAE-Net [49]. Except for MHT, CNHT, and TSCFW, the other trackers use deep features for tracking. The performance of competing methods is referenced from values provided in their original implementations. As better performance can be achieved with more features, we further extend our SSTCF tracker to SSTCF++ by further integrating the 2-D HOG feature.

B. Quantitative Comparison

1) Evaluation on the HOT2020 Dataset: We first compare all the methods on the HOT2020 dataset. λ_1 and λ_2 were set to $(0.06/N)$ and 280, respectively. As shown in Fig. 3, our SSTCF and SSTCF++ achieve very competitive performance among all the methods, with SSTCF++ ranking at the top. Due to the limitations of fixed feature extraction, CNHT performs poorly among all the methods. The local–global spatial–spectral feature extraction enables MHT to significantly outperform DeepHKCF and CNHT. TSCFW surpasses the MHT thanks to the low-rank preprocessing. Thanks to end-to-end training, BAE-Net also demonstrates favorable performance. SSTCF leverages low-rank representation to capture the global correlation among all extracted features, enhancing tracking robustness. Furthermore, the temporal constraint improves tracking continuity across successive frames, helping

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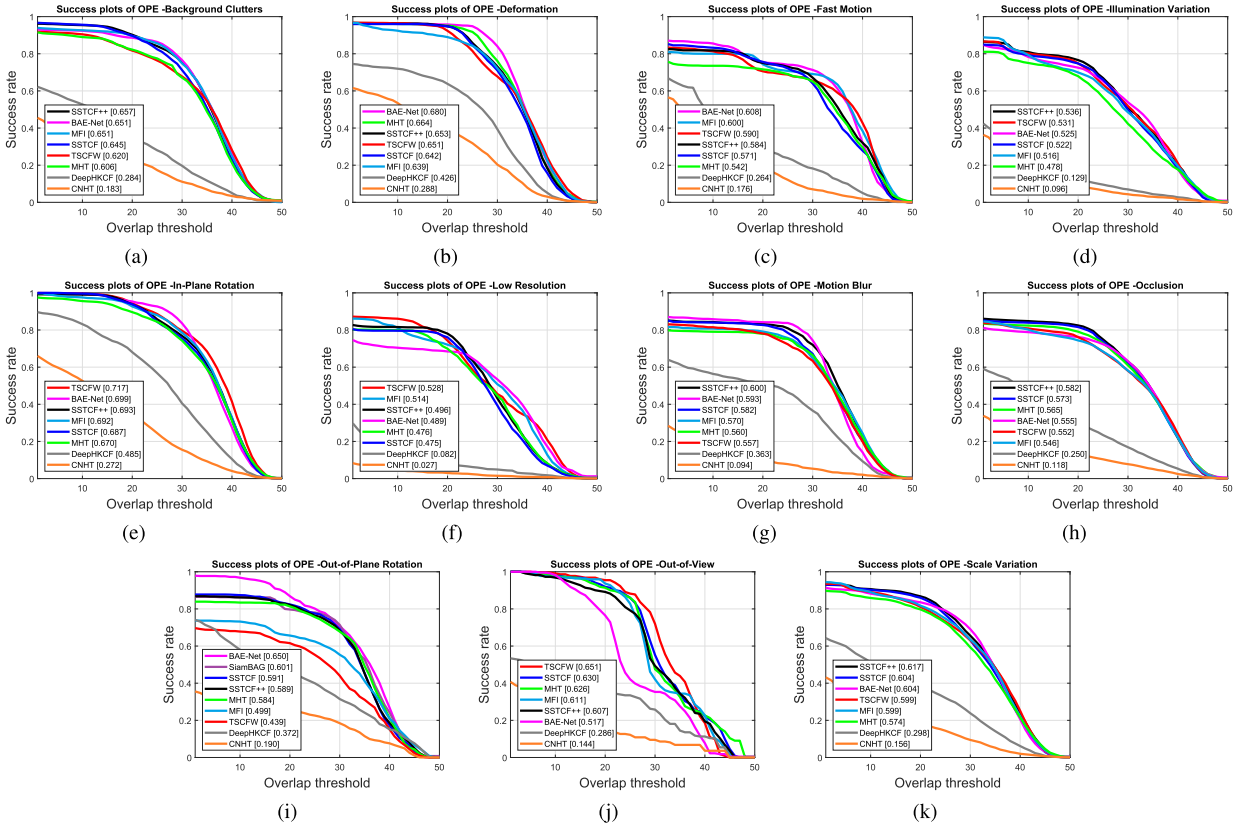


Fig. 4. Attribute-based comparison on the HOT2020 dataset. (a) BC. (b) DEF. (c) FM. (d) IV. (e) IPR. (f) LR. (g) MB. (h) OCC. (i) OPR. (j) OV. (k) SV.

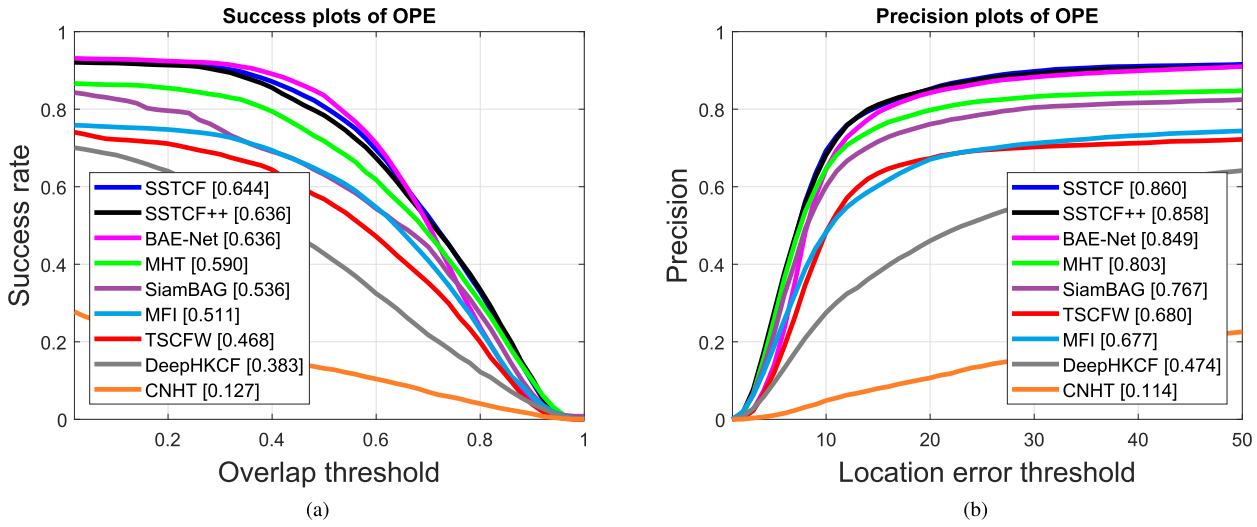


Fig. 5. Success and precision plots of all the compared hyperspectral trackers on the IMEC25 dataset. (a) Success plot. (b) Precision plot.

overcome unexpected interruptions. For these reasons, relying solely on handcrafted features for tracking, SSTCF surpasses BAE-Net. By further using the 2-D HOG, SSTCF++ even outperforms the BAE-Net to a noticeable margin with an improvement of **0.9%** in AUC and **2.5%** in DP@20. Overall, this experiment clearly demonstrates the effectiveness of our spatial–spectral–temporal tracking approach.

We further provide an attribute-based comparison of all the methods in Fig. 4. We use AUC to measure all the trackers. It can be seen that our SSTCF and SSTCF++ generally rank among the top two methods across five attributes including BC, IV, MB, OCC, and SV. Thanks to more profound utilization of the spectral information, our SSTCF++ achieves the best performance in the BC scenarios. In IV cases, where illumination changes can cause dissimilarities in the extracted features across different bands, the low-rank representation allows the CFs to be compressed, effectively overcoming this issue. In MB cases, where spatial structural information is less reliable, effective modeling from the spectral aspect ensures stable tracking performance. For OCC cases, the temporal

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Fig. 6. Visual comparison on the HOT2020 dataset in *forest2*, *toy2*, *rider1*, and *pedestrain* sequences.

constraint helps maintain tracking consistency between frames, resulting in more robust tracking.

2) Evaluation on the IMEC25 Dataset: We provide a performance comparison of all the trackers on the IMEC25 dataset in Fig. 5. λ_1 and λ_2 were set to $(0.008/N)$ and 20, respectively. We additionally include SiamBAG [31] in comparison. As the objects are relatively small, we set the cell size in the SSHMG feature extractor to 2. Overall, our SSTCF

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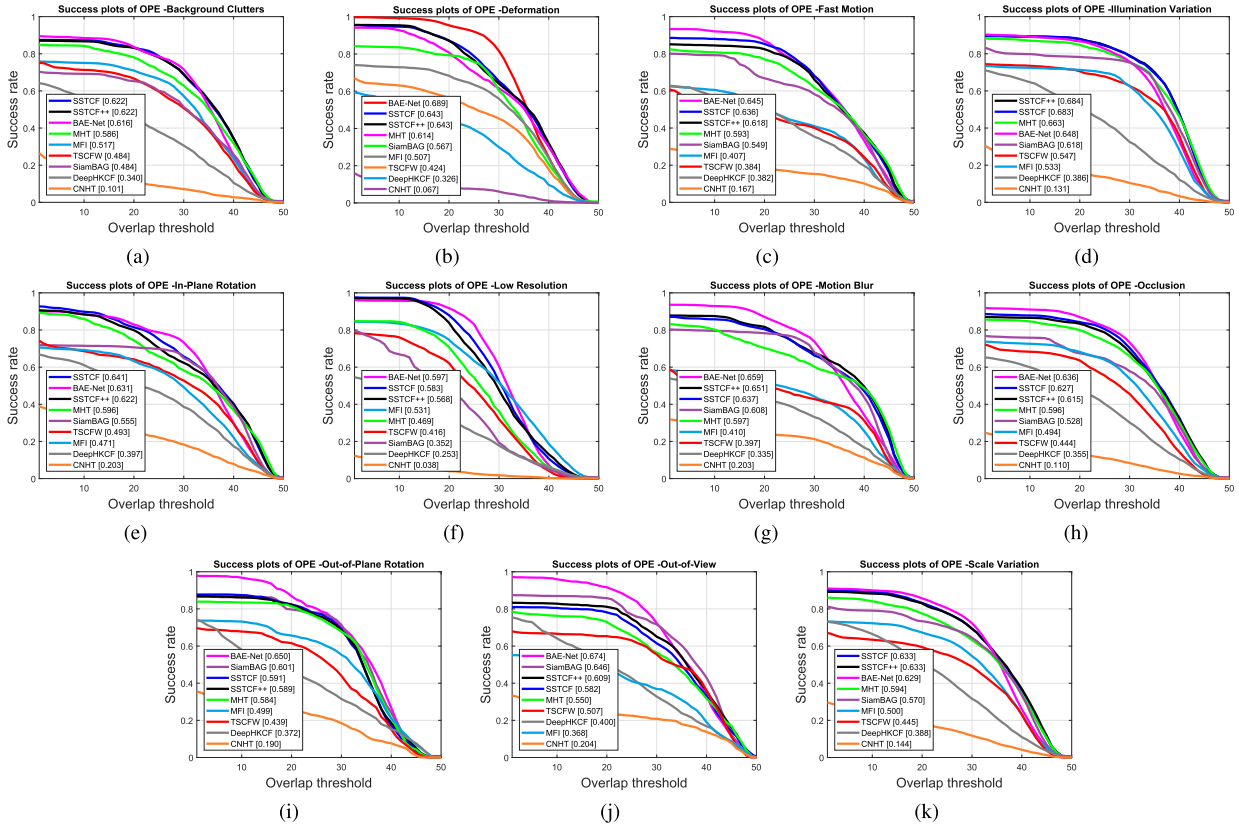


Fig. 7. Attribute-based comparison on the IMEC25 dataset. (a) BC. (b) DEF. (c) FM. (d) IV. (e) IPR. (f) LR. (g) MB. (h) OCC. (i) OPR. (j) OV. (k) SV.

TABLE I
RUNNING TIME COMPARISON OF ALL THE METHODS

Device	**GPU**		**CPU**						
Methods	BAE-Net	SiamBAG	MFI	MHT	DeepHKCF	CNHT	TSCFW	SSTCF	SSTCF++
FPS	0.72	10.52	0.42	2.23	0.91	2.61	3.90	1.05	0.99

and SSTCF++ achieve the most promising performance by providing an AUC of **0.644** and **0.636**, respectively. TSCFW and MFI show an obvious performance drop because of too many parameters to be tuned. It is interesting that the SSTCF++ did not outperform SSTCF. The possible reason is that the IMEC25 dataset contains more bands than HOT2020, usually exhibiting higher correlation. As a result, further introduction of HOG, which fails to consider the spectral correlation among bands, did not provide improvement. We further provide a visual comparison of all the methods in Fig. 6. As expected, our SSTCF and SSTCF++ still achieve very promising results.

We perform attribute-based evaluations for all trackers in Fig. 7. Our SSTCF and SSTCF++ achieve the highest performance across four attributes: BC, IV, SV, and IPR. As seen in the HOT2020 dataset, the low-rank constraint helps capture spectral correlations, leading to higher accuracy in the BC index. In addition, the temporal and low-rank constraints prove very effective in addressing illumination changes, resulting in a remarkable performance for both SSTCF++ and SSTCF in the IV index. It is also worth noting that BAE-Net achieves competitive performance in attribute-based comparisons, likely due to its robust representation capabilities.

We further record the running time of all the methods in Table I. The running configuration includes an NVIDIA GeForce RTX 3090 GPU and an Intel Xeon Silver 4214R CPU operating at 2.40 GHz. Thanks to the accelerated processing with GPUs, deep learning-based methods typically require less time. While our SSTCF and SSTCF++ methods do not show significant advantages compared to other approaches, this performance gap could be reduced by implementing them on GPUs. We leave this as part of our future work. We further record the running time of all the methods in Table I. The running configuration includes an NVIDIA GeForce RTX 3090 GPU and an Intel Xeon Silver 4214R CPU operating at 2.40 GHz. Thanks to the accelerated processing with GPUs, deep learning-based methods typically require less time. While our SSTCF and SSTCF++ methods do not show significant advantages compared to other approaches, this performance gap could be reduced by implementing them on GPUs. We leave this as part of our future work.

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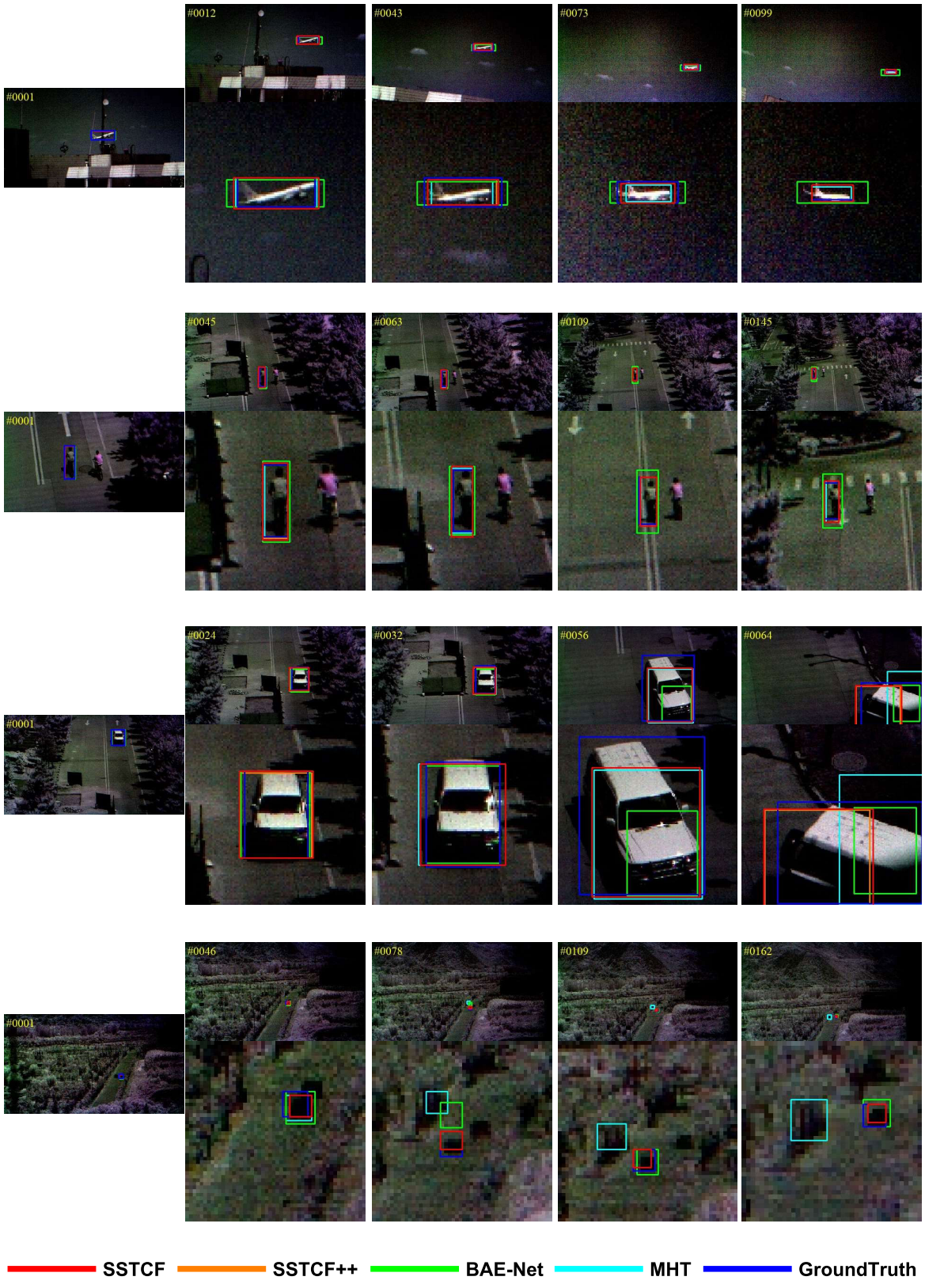


Fig. 8. Visual comparison of all the compared hyperspectral trackers on the IMEC25 dataset in *airplane6*, *car*, *doublebicycle*, and *doublecar* sequences.

We provide a visual comparison of all methods in Fig. 8. As observed, the objects appear dim, and the background is complex. In the *airplane6* sequence, the airplane has a very similar color to the clouds. Thanks to the effective utilization of spatial–spectral–temporal cues, our SSTCF and SSTCF++ successfully track the target, while other trackers either miss the object or produce inaccurate bounding boxes. In the *car* sequence, the object undergoes rotation, making the spatial

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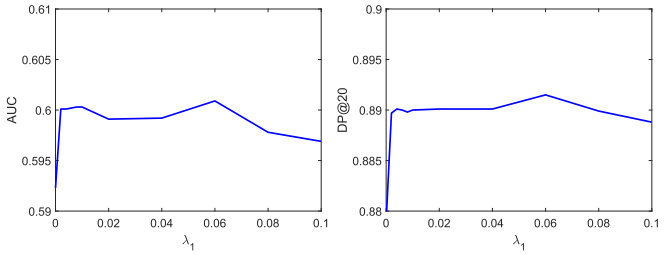


Fig. 9. Impact of the low-rank constraint. (For simplicity, N is omitted in the figure.) (a) AUC. (b) DP@20.

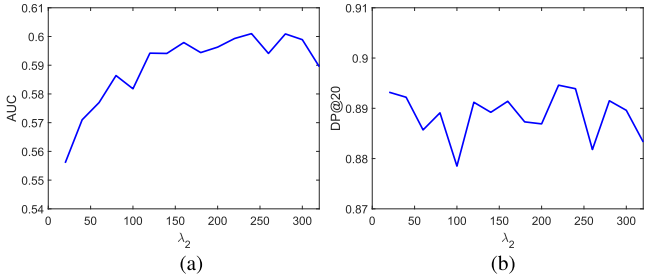


Fig. 10. Impact of the temporal constraint. (a) AUC. (b) DP@20.

TABLE II
ABLATION STUDY OF THE TEMPORAL AND LOW-RANK CONSTRAINTS ON THE HOT2020 DATASET

Temporal	Low-rank	OCC	BC	Overall
✗	✗	0.565	0.606	0.586
✓	✗	0.580	0.611	0.596
✗	✓	0.617	0.618	0.597
✓	✓	0.573	0.645	0.601

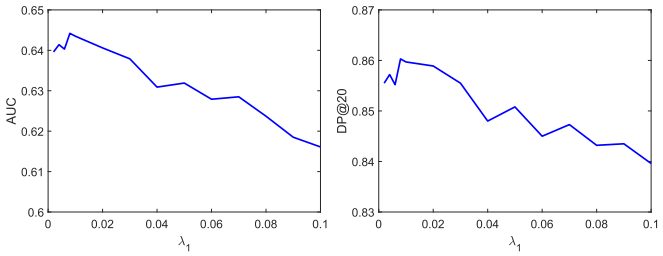


Fig. 11. Impact of the low-rank constraint on the HOT2020 dataset. (For simplicity, N is omitted in the figure.) (a) AUC. (b) DP@20.

structure less reliable. However, due to the effective modeling across spectral bands, SSTCF and SSTCF++ are more capable of accurately locating the object. In the *doublebicycle* sequence where the person moves from near to far, demonstrating scale changes, SSTCF and SSTCF++ still deliver promising performance, owing to the temporal constraint. The spatial–spectral–temporal modeling also proves beneficial for tracking small objects, as shown by the good performance in the *doublecar* sequence. Overall, the visual comparison further evidently demonstrates the strong robustness of our proposed trackers.

C. Parameter Analysis

Here, we provide an analysis of the effect of low-rank and temporal constraints on the tracking performance.

1) Experiments on the HOT2020 Dataset: The effect of the low-rank constraint in the SSTCF framework is governed by the parameter λ_1. To study its impact on tracking performance, we conduct experiments where the hyperparameter for the temporal constraint λ_2 is fixed at 280. We then vary λ_1 from $(0/N)$ to $(0.1/N)$ with increments of $(0.02/N)$ to observe the influence of the low-rank constraint. As illustrated in Fig. 9, smaller values of λ_1 do not sufficiently capture the low-rank characteristics of the filters, leading to suboptimal utilization of the extracted features for tracking. As a result, the performance gradually improves with the increasing λ_1. However, when λ_1 becomes too large, it tends to overestimate the low rankness, which negatively impacts performance. Based on these observations, the optimal value of λ_1 is determined to be $(0.06/N)$ in our experiment.

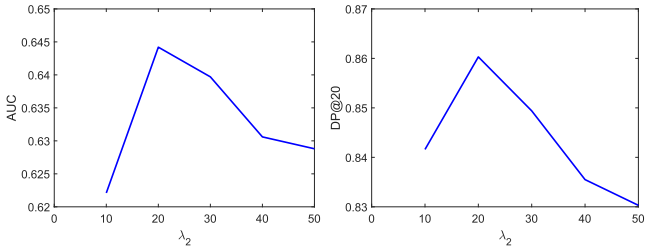
The parameter λ_2 controls the influence of the temporal constraint in the SSTCF framework, impacting how well the tracker maintains consistency across consecutive frames. Fixing $\lambda_1 = (0.06/N)$, we change λ_2 from 20 to 320 with an interval of 20. As shown in Fig. 10, when λ_2 is set to smaller values, the temporal continuity between frames is not adequately modeled, leading to potential inconsistencies in tracking. Conversely, larger values of λ_2 overemphasize temporal continuity, potentially neglecting necessary adjustments for changes in consecutive frames. Through this analysis, we find that setting $\lambda_2 = 280$ strikes a good balance between maintaining temporal continuity and allowing for adaptability to changes, resulting in optimal tracking performance.

We further show the performance improvement with/without the temporal and low-rank constraints in Table II. The performance is measured with the AUC of the success plots. The temporal constraint is helpful to ensure filter consistency across consecutive frames and improves tracking continuity between nearby frames, resulting in performance improvement in OCC scenarios. Thanks to the low-rank constraint, the global spectral semantical dependencies among learned filters are profoundly depicted, leading to a more profound utilization of the spectral information. As a result, the performance further improves in the BC scenarios.

2) Experiments on the IMEC25 Dataset: To investigate the impact of low-rank constraint on the tracking performance, we vary λ_1 while keeping the hyperparameter for the temporal constraint λ_2 fixed at 20. From Fig. 11, the performance initially increases and then decreases as λ_1 increases, reaching its peak at $\lambda_1 = (0.008/N)$. Based on these results, λ_1 is set to $\lambda_1 = (0.008/N)$ for the IMEC25 dataset. With λ_1 fixed at $(0.008/N)$, we vary λ_2 from 10 to 50 in increments of 10 to see the impact of temporal constraint. As illustrated in Fig. 12, we observe that setting $\lambda_2 = 20$ achieves the best performance.

We further demonstrate the performance improvement with and without the temporal and low-rank constraints in Table III on the IMEC25 dataset. It can be observed that both the low-rank and temporal constraints contribute positively to the performance.

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(a) (b)

Fig. 12. Impact of the temporal constraint on the IMEC25 dataset. (a) AUC. (b) DP@20.

TABLE III
ABLATION STUDY OF THE TEMPORAL AND LOW-RANK CONSTRAINTS ON THE IMEC25 DATASET

Temporal	Low-rank	OCC	BC	Overall
✗	✗	0.596	0.586	0.590
✓	✗	0.627	0.626	0.640
✗	✓	0.605	0.586	0.595
✓	✓	0.627	0.622	0.644

VI. CONCLUSION

In this article, we introduce a spatial–spectral–temporal CF framework for hyperspectral object tracking. The low-rank constraint is designed to capture the global correlation of the filter within each frame, while the temporal constraint enforces tracking consistency between frames. Experimental results demonstrate that both the low-rank and temporal constraints significantly enhance tracking robustness. In future work, we plan to extend our framework to a deep learning approach to further improve tracking accuracy and speed.

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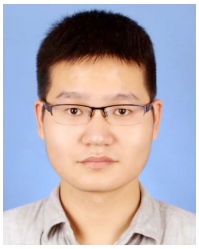
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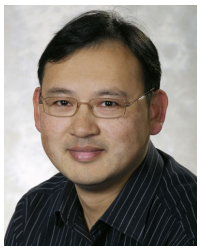
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